

Reinforcing Recursive Language Models: A Literature Review

Pablo Leyva

first-principles-to-llms chain, study notes

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Notation

We adopt the symbol set used in the companion deep dive (`02-math-deep-dive.md`). Throughout, π_θ denotes a single shared autoregressive policy that plays both root and child roles in a recursive language-model (RLM) trajectory tree.

Symbol	Meaning
$\pi_\theta(a \mid s)$	Token-level policy: probability of token a given context s .
G	Number of root rollouts per query (GRPO group size).
k_g	Number of child rollouts spawned by root g .
A_g	Group-relative advantage: $A_g = (r_g - \mu_r) / (\sigma_r + \delta)$.
$\rho_\theta(y^{(t)})$	Importance ratio at token t , $\pi_\theta / \pi_{\theta_{\text{old}}}$.
ε	PPO clipping radius (typically 0.2).
$\mathcal{T}$	Full trajectory tree (root plus descendant rollouts).

Background

A *recursive language model* (RLM) [Zhang et al., 2026] treats each LM session as a process that runs inside a Python REPL sandbox. The sandbox exposes three privileged calls: `FINAL(answer)` terminates the rollout, `rlm_query(prompt, context)` spawns a single child rollout under the same policy, and `rlm_query_batched(prompts, context)` spawns a list. Every `rlm_query` invocation grows the trajectory tree $\mathcal{T}$ by attaching a child whose own behaviour is again drawn from π_θ . Recursion turns the per-token causal-language-model MDP — in which states are prefixes and actions are next tokens [Schulman et al., 2017] — into a tree-structured trajectory whose leaves all eventually call `FINAL`.

The motivation is long-context reasoning. Modern frontier models still struggle when the relevant evidence is scattered across millions of tokens, e.g. multi-paper scientific corpora. The RLM scaffold lets a small model *decompose* a long-context query into many short-context child queries (“search $\rightarrow$ expand $\rightarrow$ extract”), each of which fits comfortably in the backbone’s effective context. Chain-of-thought prompting [Wei et al., 2022] unlocked sequential intermediate reasoning; the RLM idea is the natural next agentic-scaling move, replacing a flat thought stream with a recursive tree.

Method

The blog’s central contribution is to RL-fine-tune a *single* shared policy π_θ to play both roles, rather than freezing the children as in the original RLM setup. The optimisation primitive is GRPO [Shao

et al., 2024, DeepSeek-AI, 2025]: for each query x , sample G root rollouts $\{y_g\}_{g=1}^G \sim \pi_{\theta_{\text{old}}}(\cdot | x)$, score each with a rubric reward $r_g \in [0, 1]$ (an LLM-judge call), and form group-relative advantages

$$A_g = \frac{r_g - \mu_r}{\sigma_r + \delta}, \quad \mu_r = \frac{1}{G} \sum_g r_g, \quad \sigma_r = \text{stddev}(r_g).$$

Each rollout (root or child) contributes a token-level PPO-clipped surrogate [Schulman et al., 2017]:

$$\mathcal{L}_{\text{node}}(y, A; \theta) = \frac{1}{|y|} \sum_{t=1}^{|y|} \min\left(\rho_{\theta}(y^{(t)})A, \text{clip}(\rho_{\theta}(y^{(t)}), 1-\varepsilon, 1+\varepsilon)A\right). \quad (1)$$

The novel piece is the **advantage-inheritance** rule. Each child rollout $y_{g,i}$ inherits the parent’s GRPO advantage, $A_{g,i} := A_g$, and the per-root contribution to the loss is balanced by $1/k_g$:

$$\mathcal{L}_g^{\text{full}}(\theta) = \mathcal{L}_{\text{node}}(y_g, A_g; \theta) + \frac{1}{k_g} \sum_{i=1}^{k_g} \mathcal{L}_{\text{node}}(y_{g,i}, A_g; \theta). \quad (2)$$

Without the $1/k_g$ factor, a root that spawned more children would dominate the gradient and the optimiser would develop a perverse incentive to call `rlm.query` more often. With it, every root contributes equally regardless of how branchy its subtree is.

The depth-1 form (2) extends recursively to arbitrary depth:

$$\mathcal{L}_{\text{subtree}}(y, A; \theta) = \mathcal{L}_{\text{node}}(y, A; \theta) + \frac{1}{k_y} \sum_{i=1}^{k_y} \mathcal{L}_{\text{subtree}}(y_i, A; \theta). \quad (3)$$

Every descendant of root g carries the same advantage A_g , and the $1/k_y$ averaging at every level keeps the depth-balanced contribution invariant under the local branching factor.

Results

The Qwen3.5-4B backbone is first cold-started by SFT on teacher rollouts distilled from Qwen3.5-397B-A17B, then RL-fine-tuned with the objective above. Training runs on a single 8×H200 node with batch 16, $G = 8$ root samples per prompt and k_g up to 4, yielding up to $16 \cdot 8 \cdot 4 = 512$ concurrent rollouts per gradient step. The average rubric score on the training distribution rises from 0.30 to 0.60 over RL.

On a held-out multi-paper evidence-selection benchmark, the RL-fine-tuned 4B model reaches an average rubric score of ≈ 0.60 , matching Claude Sonnet 4.6’s 0.607 on the same scaffold while running in ≈ 7 s per query against Sonnet’s ≈ 60 s — an $8.5\times$ wall-clock speedup at roughly two orders of magnitude fewer parameters. As a quality result this is a *tie*; the production win is latency and serving cost.

Position in the literature

Original RLM paper. Zhang et al. [2026] introduced the recursive-LM scaffold and showed that a *prompted* (not RL-trained) frontier model can decompose long-context queries into a tree of sub-rollouts. They trained only the root, leaving children to a frozen LM. The blog reviewed here removes that asymmetry: one policy, both roles, end-to-end RL.

Native parallelism. Wu et al. [2025] take a similar single-policy RL stance but replace the recursion tree with a *flat* parallel-reasoner ensemble distilled from the model itself. The two papers share the ethos of training a small model to exploit its own parallelism; they differ on whether the parallelism is recursive (tree, this work) or flat (ensemble, theirs).

Pre-training / mid-training / RL interplay. Zhang et al. [2025] survey where in the modern training pipeline RL pays off. Their findings frame why RLM training works at the 4B scale: a sufficiently strong mid-trained backbone is a prerequisite for GRPO to lift rubric scores the way the blog reports.

GRPO behaviour vs SFT. Rajani et al. [2025] characterise GRPO as a “scalpel” that amplifies already-present capabilities while SFT acts as a “hammer” that overwrites behaviour. The cold-start SFT $\rightarrow$ GRPO recipe used by the blog is exactly the combination their analysis recommends: SFT to install the recursive scaffold behaviour, GRPO to optimise it.

Rubric-based rewards. The blog uses an LLM-judge rubric reward instead of a verifiable reward (which the authors report was too noisy: “F1 of selected snippets” had unusable variance). This methodology comes from Gunjal et al. [2025] and Yuan et al. [2025], both of whom argue that rubric rewards extend RL to domains where ground-truth verifiers are absent — exactly the evidence-selection setting here.

Chain-of-thought. Wei et al. [2022] unlocked sequential intermediate reasoning. Recursive LMs are arguably the next agentic-scaling story: where CoT replaces a single token prediction with a sequence of intermediate tokens, RLM replaces a single context with a tree of contexts. The training-time challenge moves from “elicit reasoning” to “credit-assign across a tree.”

DeepSeek-R1. DeepSeek-AI [2025] demonstrated GRPO-at-scale with rule-based rewards. The blog inherits the GRPO machinery but swaps rule-based rewards for rubric rewards and extends the credit-assignment to a recursive trajectory tree.

Hierarchical RL — options framework. A connection the blog does not draw explicitly: RLM children are very nearly *options* in the sense of Sutton et al. [1999]. An option is a tuple (I, π_o, β) where I is the initiation set, π_o the option policy, and β the termination condition. For an RLM child, I is the REPL state at the moment of `rlm.query`, $\pi_o = \pi_\theta$ (the shared policy — so options share parameters), and β fires precisely when the child calls `FINAL`. Read this way, the recursive-subtree loss (3) is a temporally-abstract policy gradient that backs up the *root-level* advantage through every option in the tree; the $1/k_g$ averaging is what re-weights options so their density does not bias the root gradient. Importing convergence and unbiasedness theorems from the options literature is, in our view, the cleanest theoretical hardening of the blog’s informal unbiasedness argument.

Engineering substrate. The implementation lives in NovaSky-AI Team [2025] (specifically PR #1596), an extension of the SkyRL framework for recursive and agentic LM training.

Discussion

Well-supported. The core empirical claim — that a single shared policy with parent-advantage inheritance and $1/k_g$ averaging trains a 4B RLM to Sonnet-4.6 parity on evidence selection — is well-supported by the reported numbers. The $1/k_g$ balancing and the cold-start SFT recipe are both standard moves in the GRPO literature [Shao et al., 2024, DeepSeek-AI, 2025, Rajani et al., 2025] and inherit those papers’ training-stability story.

Hand-waved. The unbiasedness of advantage inheritance is argued informally. Inheriting A_g for every descendant is unbiased only when the child’s reward contribution is θ -independent given the parent’s rollout window; this holds for evidence selection (parent reward is roughly additive in child contributions) but fails for tasks with non-additive child interactions (e.g. tool-use chains where child A’s output is child B’s input). The blog flags “more fine-grained credit assignment” as future work but does not formalise the load-bearing assumption. Likewise, the model still slightly trails Sonnet 4.6 on rubric quality (0.60 vs 0.607); the headline result is wall-clock parity, not quality breakthrough.

Future directions. Three open questions, expanded in the companion `05-improvements.tex`: (i) does the recursive-subtree loss (3) train at depth $d > 1$? The blog only trains depth-1; (ii) at fixed compute, what is the optimal k_g , and is there a knee in the rubric-vs- k_g curve? (iii) can RL discover the search-expand-extract strategy from scratch rather than being prompted into it?

See also: cross-links to the first-principles-to-llms chain

This paper sits naturally on top of several chapters of the chain repository [first-principles-to-llms](#):

- Causal-LM-as-MDP framing ([Ch 25](#)) — recovered at every node of the trajectory tree.
- Tiny-GPT pre-training ([Ch 27](#)) — the backbone scale at which a GRPO-trained model can match a much larger frontier model.
- SFT, RLHF, DPO ([Ch 28](#)) — the cold-start SFT step that precedes RL here.
- MDP foundations ([Ch 29](#)) — where the per-token MDP and value/advantage definitions live.
- Maximum-entropy RL ([Ch 30](#)) — conceptual prior for the PPO clipping/KL machinery.
- Policy gradient and the GRPO derivation ([Ch 31](#)) — the surrogate (1) and the group-baseline variance-reduction argument that the blog inherits.

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